Prometheus

Autonomous ML Pipeline Builder — Test Prompts & Datasets

Use these prompts and CSV links to test all four task types. Download each CSV first then upload to Prometheus.

Binary Classification

TEST 1 — Titanic Survival

CSV:

```
https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv
```

Prompt:

```
Predict whether a passenger survived the Titanic disaster. Target column is Survived
(1 = survived, 0 = died).
```

Approval screen: Binary Classification · Survived · ROC-AUC

Expected: ROC-AUC above 0.80 · predictions return 0 or 1 with probability

TEST 2 — Heart Disease

CSV:

```
https://raw.githubusercontent.com/sharmaroshan/Heart-UCI-Dataset/master/heart.csv
```

Prompt:

```
Predict whether a patient has heart disease based on clinical measurements. Target
column is target (1 = disease, 0 = no disease).
```

Approval screen: Binary Classification · target · ROC-AUC

Expected: Accuracy above 85% · confidence above 80% on predictions

TEST 3 — Customer Churn (Yes/No encoding test)

CSV:

```
https://raw.githubusercontent.com/IBM/telco-customer-churn-on-icp4d/master/data/Telc
o-Customer-Churn.csv
```

Prompt:

```
Predict whether a telecom customer will churn and cancel their subscription. Target
column is Churn.
```

Approval screen: Binary Classification · Churn · ROC-AUC

Expected: Churn column correctly encoded · predictions return Yes/No not 0/1

Regression

TEST 4 — California Housing

CSV:

```
https://raw.githubusercontent.com/ageron/handson-ml/master/datasets/housing/housing.
csv
```

Prompt:

```
Predict the median house value for California districts based on demographic and
geographic features. Target column is median_house_value.
```

Approval screen: Regression · median_house_value · RMSE

Expected: RMSE under $60,000 · predictions return dollar amounts · R2 above 0.75

Multi-Class Classification

TEST 5 — Iris Species (3 classes)

CSV:

```
https://raw.githubusercontent.com/mwaskom/seaborn-data/master/iris.csv
```

Prompt:

```
Classify iris flowers into one of three species based on petal and sepal
measurements. Target column is species.
```

Approval screen: Multiclass Classification · species · F1 Macro

Expected: F1 Macro above 0.95 · all 3 class probabilities shown · setosa at 100% confidence

TEST 6 — Wine Quality (3 classes, harder)

CSV:

```
https://raw.githubusercontent.com/dsrscientist/dataset1/master/winequality-red.csv
```

Prompt:

```
Classify red wine quality as poor, average, or good based on chemical properties.
Target column is quality. Bin scores: 1-4=poor, 5-6=average, 7-10=good.
```

Approval screen: Multiclass Classification · quality · F1 Macro

Expected: F1 Macro 0.70-0.80 · tests class imbalance handling

Time Series Forecasting

TEST 7 — Air Passengers (simplest)

CSV:

```
https://raw.githubusercontent.com/jbrownlee/Datasets/master/airline-passengers.csv
```

Prompt:

```
Forecast monthly airline passenger numbers for the next 12 months. Date column is
Month, target column is Passengers.
```

Approval screen: Time Series · Month · Passengers · Horizon: 12 · MAPE

Expected: MAPE under 15% · forecast chart shows blue/green/orange lines · 12-month table with CSV download

TEST 8 — Daily Temperature

CSV:

```
https://raw.githubusercontent.com/jbrownlee/Datasets/master/daily-min-temperatures.csv
```

Prompt:

```
Forecast daily minimum temperature for the next 30 days. Date column is Date, target column is Temp.
```

Approval screen: Time Series · Date · Temp · Horizon: 30 · MAPE

Expected: MAPE under 20% · tests daily frequency detection

TEST 9 — Monthly Car Sales (trend + seasonality)

CSV:

```
https://raw.githubusercontent.com/jbrownlee/Datasets/master/monthly-car-sales.csv
```

Prompt:

```
Forecast monthly car sales for the next 6 months. Date column is Month, target column is Sales.
```

Approval screen: Time Series · Month · Sales · Horizon: 6 · MAPE

Expected: MAPE under 25% · tests trend + seasonality detection

Sanity Checks

Check	Binary	Regression	Multiclass	Time Series
Metric shown	ROC-AUC / Accuracy	RMSE / R2	F1 Macro	MAPE / RMSE
Prediction format	0/1 or Yes/No	Dollar amount	Class name + all probs	Future values + chart
Should NOT show	RMSE or dollar amounts	ROC-AUC or Yes/No	Single confidence only	Yes/No or 0/1
Health check	classification: healthy	regression: healthy	multiclassification: healthy	timeseries: healthy

Gateway Health Check

```
curl http://localhost:8000/health
```

Expected response: {"classification": "healthy", "regression": "healthy", "multiclassification": "healthy", "timeseries": "healthy"}